Md. Yousuf

Applicant for fully funded Ph.D. programs in Computer Science · Fall 2027

Medical image analysis · Deep learning · Computer vision · Semantic segmentation

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GitHub | LinkedIn | IEEE Xplore 11545645 | IUB thesis record

RESEARCH INTERESTS

I am applying for a fully funded Ph.D. in medical image analysis, computer vision, or deep learning. My undergraduate research built a thoracic CT segmentation pipeline (Hounsfield-unit preprocessing, U-Net with a ResNet-18 encoder, PyTorch Lightning, Dice/IoU). I want to extend this toward 3D models, attention, limited-label learning, and methods that hold up across scanners and institutions.

EDUCATION

Independent University, Bangladesh (IUB) · B.Sc. in Computer Science and Engineering · *Minor in Management Information Systems*

CGPA 3.43 / 4.00 · Thesis term Spring 2026 · IUB repository record dated April 2026

Thesis. *Deep Learning Framework for Automated Lung and Pulmonary Nodule Segmentation in Thoracic CT Scans.* Supervisor: Md. Rashedur Rahman, D. Eng., Assistant Professor, Department of Computer Science and Engineering, School of Engineering, Technology and Sciences. Co-authors: Abbas Yasin and Muaaz Mohammad.

Relevant study. AI, data structures and algorithms, theory of computation, discrete mathematics, numerical methods, calculus, databases, operating systems, networks, computer architecture, compilers, embedded systems. Pre-university studies: Milestone College, Uttara.

PUBLICATIONS

IEEE paper and IUB thesis are related but separate; metrics are quoted from each source and not mixed.

[1] M. Yousuf, A. Yasin, M. Mohammad, F. Israq, S. T. Bhuiyan, R. Rahman, S. B. Alam, and A. Islam, *"Deep Learning Framework for Automated Pulmonary Nodule Identification and Anatomical Segmentation in Thoracic CT Scans,"* in Proc. 2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN), 2026, pp. 1-6. doi: 10.1109/qpain69676.2026.11545645.

[2] M. Yousuf, A. Yasin, and M. Mohammad, *"Deep Learning Framework for Automated Lung and Pulmonary Nodule Segmentation in Thoracic CT Scans,"* undergraduate thesis, Independent University, Bangladesh, Spring 2026. https://ar.iub.edu.bd/handle/11348/1204

RESEARCH EXPERIENCE

Undergraduate thesis research | *Independent University, Bangladesh* | Supervisor: Md. Rashedur Rahman, D. Eng.

- Built a four-class U-Net with a ResNet-18 encoder (background, left lung, right lung, pulmonary nodules) in PyTorch Lightning, trained with Dice loss, Adam, a cosine learning-rate schedule, and GPU compute.
- CT preprocessing: Hounsfield unit window centered at -600 (width 1500); slices resized to 224 x 224 grayscale, with channel replication when required. Datasets in the described workflow: LUNA16 and LIDC-IDRI.
- IUB thesis abstract: Dice 0.996 (left lung), 0.973 (right lung), 0.848 (nodules). Limitations noted there: limited training data and 2D slice-based processing.
- IEEE QPAIN abstract: 888 LUNA16 cases; mean IoU 0.994 / 0.960 / 0.741 (left lung, right lung, nodules); aggregate Dice 0.947, with cross-subset validation reported as stable.

TEACHING AND ACADEMIC SUPPORT

Teacher Assistant | *Introduction to Computer Science, Independent University, Bangladesh*

- Teaching assistant for Introduction to Computer Science at IUB.

Personal Assistant | *Prof. Yusuf Mahbubul Islam, PhD* | Vice Chancellor and Professor, Southeast University ·

approx. 2 years

- Personal assistant for approximately two years while foundation-course work was supervised by Prof. Islam. Conducted 40+ teacher-training workshops.

Course Developer and Supervisor | *BIIT Foundation Skill Development Certification Program* | Supervisor: Prof. Yusuf Mahbubul Islam, PhD

- Reviewed lesson plans, teacher guides, slides, and practical exercises; checked alignment of objectives, content, and assessments; and coordinated QBL curriculum feedback.

PROFESSIONAL EXPERIENCE

Web and System Developer | *BIIT* | From 1 October 2026

- Website development, system implementation, and related technical work.

Senior Web Developer | *Bosttopus.com* | From 10 November 2025 · Remote

- Senior web developer role, remote. Project-level claims omitted until they can be verified.

SELECTED PROJECTS

Noor Al Salam Hotel. Live public hotel website with rooms, facilities, gallery, contact, and booking search. nooralsalamhotel.com. PostgreSQL specified for connected operations work.

Tasfiq property management system. Live property, office, and finance workspace (React, Vite, Tailwind CSS, Node.js, Express, MongoDB, Firebase Hosting). tasfiqpms.web.app. Inner modules not shown on the login screen are not claimed here.

Education-technology work in progress. Bridge Institute / UIT website concept and a QBL learning-platform concept.on going.

TECHNICAL SKILLS

Languages. Python, JavaScript, Java, C.
Deep learning. PyTorch, PyTorch Lightning, U-Net, ResNet-18, Dice loss, transfer learning, medical image segmentation, CT preprocessing, computer vision.
Web and systems. React.js, Tailwind CSS, Vite, Node.js, Express, REST APIs, MongoDB, PostgreSQL, Firebase, Git, GitHub, Vercel, Netlify.
Research practice. Technical documentation, research writing, curriculum development, Question-Based Learning, project planning, team coordination.

LEADERSHIP AND COMMUNITY SERVICE

Street lighting in Akanagor | *Manus Kahon* | Bancharampur, Brahmanbaria

- Addressed village darkness in Akanagor with Manus Kahon (slogan: CREATE YOUR SELF AND MAKE THE WORLD TO SMILE) through a mindmap, local fundraising, shop support, and field implementation.

Venue monitoring | *IUB Live in Field Experience* | Sreemangal

- Student communication, coordination with other monitors, workshop support, and community activities in Sreemangal.

Also. Winter clothing distribution; local medical-treatment support (no private details); clean-village environmental work.

REFERENCES

Md. Rashedur Rahman, D. Eng. · Assistant Professor, CSE, School of Engineering, Technology and Sciences, IUB · Thesis supervisor · rashed@iub.edu.bd

Prof. Yusuf Mahbubul Islam, PhD · Vice Chancellor and Professor, Southeast University · Foundation-course and personal-assistant supervisor · ymislam@seu.edu.bd